MSc Final Exam: AI and Machine Learning (Lectures 1–5)

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Instructions

This paper covers Lectures 1–5: Introduction to AI, What is Machine Learning, an ML model from scratch, Bayesian and Naive Bayes classifiers, and KNN and centroid classifiers. Every question is followed by a model answer for the examiner.

Section	Type	Questions	Marks
A	Short answer	10 × 2	20
B	Descriptive	5 × 8	40
C	Numerical problems	5 × 8	40
D	MCQ (optional practice / viva)	15 × 1	15

- Total for the written paper (A + B + C): 100 marks, 3 hours.
- Show all steps in Section C. Round to 4 decimal places unless told otherwise.
- Calculators allowed.

Section A: Short answer (10 × 2 = 20 marks)

A1. Define Artificial Intelligence and an intelligent agent.

Answer: AI is the study and design of intelligent agents. An intelligent agent is a system that perceives its environment and takes actions in it.

A2. What is the Turing Test, and who proposed it?

Answer: Proposed by Alan Turing in 1950, it tests "acting humanly". An interrogator questions a hidden human and a hidden machine. If the interrogator cannot tell which is the machine, the machine is considered intelligent.

A3. Distinguish AI, Machine Learning and Deep Learning.

Answer: AI is any technique that lets machines mimic human behaviour. ML is a subset of AI that uses statistical methods so machines learn and improve with experience. DL is a subset of ML whose algorithms (deep neural networks) let the system train itself on raw

data. Data Science overlaps all three.

A4. What is the "AI Winter"?

Answer: A period of false starts and dead ends when funding and interest in AI collapsed. The main one followed the expert-systems bust of about 1988–93.

A5. Define Machine Learning in one sentence.

Answer: Given a problem, ML builds an appropriate (training) dataset, computes on it to learn the required rules, patterns and relations, and outputs a decision or inference for new data.

A6. Differentiate classification and regression.

Answer: Both are supervised. Classification outputs one of a predefined, discrete set of labels (CAR/BIKE). Regression outputs a continuous real value (a price such as 20500.50).

A7. Name two unsupervised learning tasks.

Answer: Clustering (grouping similar objects without labels) and association rule mining (finding rules such as "customers who buy X also buy Y").

A8. What is the Vector Space Model?

Answer: Each sample is represented as a vector of feature values $v_i = \{w_{i_1}, w_{i_2}, \ldots, w_{ik}\}$, where w_{ij} is the value of feature f_j. Rows are feature vectors; columns are features.

A9. State the "naive" assumption of the Naive Bayes classifier.

Answer: The features are conditionally independent given the class, so $Pr(w_1,\ldots,w_k \mid c) \approx \Pi Pr(w_j \mid c)$.

A10. In KNN, how are neighbours ranked when using cosine similarity versus L2 distance?

Answer: With cosine similarity, sort in descending order (largest similarity first). With L2 distance, sort in ascending order (smallest distance first). Then take the top k and use majority voting.

Section B: Descriptive ($5 \times 8 = 40$ marks)

B1. Trace the history of AI from the Turing Test to large language models. Mention at least six milestones. (8)

Model answer:

Year / period	Milestone
1950	Turing proposes the Turing Test for machine intelligence
1955–56	John McCarthy coins "artificial intelligence"
1961	Unimate, first industrial robot, at GM
1964–66	ELIZA chatbot (MIT); Shakey, first general-purpose mobile robot (Stanford)
1970–90	Knowledge-based and expert systems boom; expert-systems bust (1988–93) leads to the AI Winter
1990–2012	Statistical approaches; probability and uncertainty; 1997 Deep Blue beats Kasparov; 2011 Watson wins Jeopardy, Siri launches
2012–22	Big data, deep learning, generative models; 2017 AlphaGo beats Ke Jie
2022–present	LLMs (ChatGPT), scaling laws, large reasoning models

Marks: 1 per correct milestone (up to 6) + 2 for showing the overall phases (symbolic → knowledge-based → statistical → deep learning → LLMs).

B2. Explain the three machine learning paradigms with one example each. (8)

Model answer:

- **Supervised learning:** learns a function f from a labelled training dataset (samples + ground-truth labels) and applies it to a new sample. Classification (car vs bike image) when outputs are discrete; regression (house price) when outputs are continuous.
- **Unsupervised learning:** no labels or ground truth. The task is to find patterns: clustering similar objects, or mining association rules (market-basket analysis).
- **Reinforcement learning:** learning by trial and error. An agent acts in an environment, receives a reward or punishment, and improves its policy. Example: a baby learning to walk, or a game-playing agent.

B3. Describe the steps to build a vehicle classifier (car vs bike) from scratch. Why is colour a bad feature? (8)

Model answer:

1. Identify features that represent the objects: F = {#Wheel, Height, Weight, Colour}.

2. Represent each vehicle as a feature vector (Vector Space Model), e.g. (4, 6, 500, Red).
3. Remove non-informative features. Colour appears equally in both classes (red, blue and yellow cars and bikes), so it carries no information about the class. #Wheel, Height and Weight separate the classes.
4. Build the model from data, e.g. estimate Pr(class | #Wheel) or set a threshold rule "if #Wheel < 3 then BIKE".
5. Classify a new sample by picking the class with the highest probability.

If one feature is not sufficient (a 4-wheeled bike or 2-wheeled car appears), add features such as Height (H/L) and estimate Pr(class | #Wheel, Height), or build a decision tree on both.

B4. Derive Bayes' rule for classification and explain why Naive Bayes is needed. (8)

Model answer: By definition of conditional probability:

```
\Pr(c_i \mid x) = \frac{\Pr(c_i, x)}{\Pr(x)} = \frac{\Pr(x \mid
c_i)\,\Pr(c_i)}{\sum_{j=1}^{k} \Pr(x \mid c_j)\,\Pr(c_j)}
```

The denominator is obtained by marginalising over the class label C. The predicted class is arg max over c_i of $\Pr(c_i \mid x)$. Since $\Pr(x)$ is the same for all classes, we only compare $\Pr(x \mid c_i) \times \Pr(c_i)$.

With $x = \{w_1, \ldots, w_k\}$, the full likelihood by the chain rule is $\Pr(w_1 \mid w_2,\ldots,w_k,c) \times \Pr(w_2 \mid w_3,\ldots,w_k,c) \times \ldots \times \Pr(w_k \mid c)$. Estimating these joint conditionals needs huge data and is very expensive. Naive Bayes assumes features are conditionally independent given the class:

```
\Pr(c_i \mid x) \propto \Pr(c_i) \prod_{j=1}^{k} \Pr(w_j \mid c_i)
```

Each factor is a simple count, so training is fast and needs little data.

B5. Explain the KNN and centroid-based (Rocchio) classifiers. Compare cosine similarity and L2 distance. (8)

Model answer:

- **KNN:** compute the similarity (or distance) between the new sample and every training sample, sort, select the top k, and assign the majority class. It is lazy: no training, all work at prediction time.
- **Centroid (Rocchio):** compute one centroid per class, $c = (1/|c|)\ \Sigma\ d_i$ over the samples in that class. Assign the new sample to the class whose centroid is nearest (or most similar). Fast at prediction time: one comparison per class.

Aspect	Cosine similarity	L2 (Euclidean) distance
Formula	$(A \cdot B) / (\|A\| \, \|B\|)$	$\sqrt{\Sigma(a_i - b_i)^2}$
Measures	Angle / direction	Absolute separation in feature space
Sort order	Descending (largest first)	Ascending (smallest first)
Weakness	Values bunch near 1 for positive physical features (height, weight)	Sensitive to feature scale; needs normalisation

Feature scaling and the choice of measure must be considered carefully; the two can rank neighbours differently.

Section C: Numerical problems ($5 \times 8 = 40$ marks)

C1. Bayesian classifier, Naive Bayes and Laplace smoothing. (8)

Training data:

#Wheel	Height	Class
4	H	CAR
4	H	CAR
4	H	CAR
2	L	BIKE
2	L	BIKE
2	L	BIKE
4	L	BIKE
2	H	CAR

(a) Using Bayes' rule (no independence assumption), compute Pr(CAR | {4, H}) and Pr(BIKE | {4, H}). (3) (b) Classify {4, H} with Naive Bayes. (2) (c) Height can also take the value M (medium), which never appears in the data. Classify {4, M} with Naive Bayes, first without and then with Laplace smoothing (K = 1). (3)

Solution:

(a) Pr(CAR) = Pr(BIKE) = 4/8 = 0.5. Pr({4,H} | CAR) = 3/4 = 0.75; Pr({4,H} | BIKE) = 0. Pr({4,H}) = 0.75 × 0.5 + 0 × 0.5 = 0.375. Pr(CAR | {4,H}) = 0.375 / 0.375 = **1**; Pr(BIKE | {4,H}) = **0**.

(b) Pr(4 | CAR) = 3/4, Pr(H | CAR) = 4/4, Pr(4 | BIKE) = 1/4, Pr(H | BIKE) = 0/4. CAR: 0.75 × 1 × 0.5 = 0.375. BIKE: 0.25 × 0 × 0.5 = 0. **Class = CAR.**

(c) Without smoothing, Pr(M | CAR) = Pr(M | BIKE) = 0, so both scores are 0 and no decision is possible. One unseen value wipes out the whole product.

Laplace smoothing:

```
P_{LAP,K}(x \mid y) = \frac{c(x,y) + K}{c(y) + K\,|X|}
```

c(x,y) = count of x in class y, c(y) = count of class y, |X| = number of distinct values of the feature. Here |Height| = |{H, M, L}| = 3 and |#Wheel| = |{2, 4}| = 2.

- Pr(M | CAR) = (0 + 1)/(4 + 3) = 1/7; Pr(4 | CAR) = (3 + 1)/(4 + 2) = 4/6
- Pr(M | BIKE) = (0 + 1)/(4 + 3) = 1/7; Pr(4 | BIKE) = (1 + 1)/(4 + 2) = 2/6
- CAR: 0.5 × 4/6 × 1/7 = 0.0476. BIKE: 0.5 × 2/6 × 1/7 = 0.0238. **Class = CAR.**

C2. Naive Bayes on a species dataset. (8)

No	Colour	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

Classify the new instance (Colour = Green, Legs = 2, Height = Tall, Smelly = No).

Solution: P(M) = P(H) = 4/8 = 0.5.

Likelihood	M	H
P(Green | class)	2/4	1/4
P(Legs = 2 | class)	1/4	4/4
P(Tall | class)	1/4	2/4

Likelihood	M	H
P(Smelly = No | class)	1/4	3/4

- Score(M) = 0.5 × 2/4 × 1/4 × 1/4 × 1/4 = 0.0039
- Score(H) = 0.5 × 1/4 × 4/4 × 2/4 × 3/4 = 0.0469
- Score(H) > Score(M), so the instance is **species H**.

Examiner's note: the lecture slide lists P(Tall | M) = 3/4 and P(Short | M) = 1/4, giving 0.0117. Counting rows 1–4 gives Tall = 1/4 (only row 2), so 0.0039 is correct. Both versions predict H; accept either working.

C3. KNN on Iris data, k = 3. (8)

Sample	Sepal L (cm)	Sepal W (cm)	Petal L (cm)	Class
d1	5.1	3.5	1.4	Setosa
d2	5.4	3.7	1.5	Setosa
d3	5.4	3.4	1.7	Setosa
d4	7.0	3.2	4.7	Versicolor
d5	5.0	2.0	3.5	Versicolor
d6	5.9	3.2	4.8	Versicolor
d7	6.3	3.3	6.0	Virginica
d8	6.5	3.2	5.1	Virginica
d9	6.9	3.2	5.7	Virginica

Classify d_new = (6.0, 3.4, 4.5) using (a) cosine similarity and (b) L2 distance. (c) Do both measures select the same neighbours?

Solution: ‖d_new‖ = √(36 + 11.56 + 20.25) = √67.81 = 8.2347.

Sample	Class	Cosine ω	Cosine rank	L2 distance	L2 rank
d6	Versicolor	0.9990	1	0.3742	1
d8	Virginica	0.9982	2	0.8062	2
d4	Versicolor	0.9969	3	1.0392	3
d9	Virginica	0.9957	4	1.5133	4
d5	Versicolor	0.9936	5	1.9900	6

Sample	Class	Cosine ω	Cosine rank	L2 distance	L2 rank
d7	Virginica	0.9921	6	1.5330	5
d3	Setosa	0.9491	7	2.8636	7
d2	Setosa	0.9354	8	3.0741	8
d1	Setosa	0.9344	9	3.2296	9

Example working: ω(d6) = (5.9·6.0 + 3.2·3.4 + 4.8·4.5) / (√68.09 × 8.2347) = 67.48 / 67.5503 = 0.9990. D(d6) = √(0.01 + 0.04 + 0.09) = √0.14 = 0.3742.

(a) Top 3 by cosine: d6, d8, d4 → Versicolor 2, Virginica 1 → **Versicolor**. (b) Top 3 by L2: d6, d8, d4 → **Versicolor**. (c) Yes, both select exactly {d6, d8, d4}. The rankings differ only further down (d5 and d7 swap at ranks 5–6).

C4. KNN-based malware classification, k = 5. (8)

Features: x = (Cache, CPU, Memory) instruction counts.

Sample	Cache	CPU	Memory	Class
d1	50	19	90	Malware
d2	51	20	91	Malware
d3	47	18	89	Malware
d4	50	21	91	Malware
d5	31	17	50	Benign
d6	30	11	40	Benign
d7	25	9	60	Benign
d8	20	10	55	Benign

Classify d_new = (65, 15, 80) using (a) L2 distance and (b) cosine similarity. Compare the results.

Solution:

Sample	Class	Squared diffs (C, P, M)	L2 distance	L2 rank	Cosine ω	Cosine rank
d1	Malware	225 + 16 + 100 = 341	18.4662	1	0.9844	3
d2	Malware	196 + 25 + 121 = 342	18.4932	2	0.9847	2
d4	Malware	225 + 36 + 121 = 382	19.5448	3	0.9828	5
d3	Malware	324 + 9 + 81 = 414	20.3470	4	0.9807	6

Sample	Class	Squared diffs (C, P, M)	L2 distance	L2 rank	Cosine ω	Cosine rank
d7	Benign	1600 + 36 + 400 = 2036	45.1221	5	0.9597	7
d5	Benign	1156 + 4 + 900 = 2060	45.3872	6	0.9830	4
d8	Benign	2025 + 25 + 625 = 2675	51.7204	7	0.9459	8
d6	Benign	1225 + 16 + 1600 = 2841	53.3010	8	0.9967	1

(a) L2, 5 nearest: d1, d2, d4, d3, d7 → Malware 4, Benign 1 → **Malware**. (b) Cosine, top 5: d6, d2, d1, d5, d4 → Malware 3, Benign 2 → **Malware**. (c) Both predict Malware, but cosine ranks benign d6 first: (30, 11, 40) points in nearly the same direction as (65, 15, 80) despite being far away in absolute terms. Cosine ignores magnitude, so its vote is much closer (3:2 vs 4:1).

C5. Centroid-based (Rocchio) classifier. (8)

Sample	Height (cm)	Weight (kg)	Class
d1	167	51	Underweight
d2	182	62	Normal
d3	176	69	Normal
d4	173	64	Normal
d5	172	65	Normal
d6	174	56	Underweight
d7	169	58	Normal
d8	173	57	Normal
d9	170	55	Normal

(a) Compute the class centroids. (3) (b) Classify d_new = (170, 57) by the nearest centroid using L2 distance and using cosine similarity. (3) (c) KNN (k = 5) classifies this sample as Normal with both measures. Comment on the difference. (2)

Solution:

(a) c_Underweight = ((167 + 174)/2, (51 + 56)/2) = (170.5, 53.5). c_Normal = (1215/7, 430/7) = (173.571, 61.429).

(b) L2: D(c_U) = √(0.25 + 12.25) = 3.5355; D(c_N) = √(12.755 + 19.612) = 5.6892. Nearest is **Underweight**. Cosine: ω(c_U) = 0.999811; ω(c_N) = 0.999862. Most similar is **Normal**.

(c) KNN looks at individual nearby samples, and 7 of 9 training samples are Normal, so the local majority is Normal. The centroid method summarises each class by its mean; the Underweight mean (170.5, 53.5) happens to lie closer in Euclidean terms. Cosine values are all near 1 for these positive physical features, so it barely separates the classes. Lesson: the choice of classifier and measure, plus feature scaling, can change the decision.

Section D: MCQs ($15 \times 1 = 15$ marks)

1. The Turing Test evaluates a machine's ability to: (a) think rationally (b) act humanly (c) learn from data (d) optimise a goal
2. The term "artificial intelligence" was coined by: (a) Alan Turing (b) Joseph Weizenbaum (c) John McCarthy (d) Garry Kasparov
3. Which system beat world chess champion Garry Kasparov in 1997? (a) Watson (b) AlphaGo (c) Deep Blue (d) Shakey
4. Deep Learning is a subset of: (a) Data Science only (b) Machine Learning (c) Robotics (d) Expert systems
5. Predicting a car's price in 2025 is an example of: (a) classification (b) clustering (c) regression (d) reinforcement learning
6. A classifier trained on CAR/BIKE is given a bus. It will output: (a) BUS (b) nothing (c) either CAR or BIKE (d) an error
7. Learning by trial and error with reward and punishment is: (a) supervised (b) unsupervised (c) reinforcement (d) semi-supervised
8. Association rule mining belongs to: (a) supervised learning (b) unsupervised learning (c) reinforcement learning (d) regression
9. In the vehicle example, which feature is non-informative? (a) #Wheel (b) Height (c) Weight (d) Colour
10. In Bayes' rule, Pr(x) in the denominator is computed by: (a) marginalising over the class labels (b) the chain rule (c) Laplace smoothing (d) majority voting
11. Naive Bayes assumes the features are: (a) normally distributed (b) conditionally independent given the class (c) equally important (d) continuous
12. Laplace smoothing is used to: (a) normalise features (b) avoid zero likelihood estimates (c) reduce k in KNN (d) compute centroids
13. With K = 1, |CAR| = 4 and Height $\in$ {H, M, L}, P_LAP(M | CAR) for an unseen M is: (a) 0 (b) 1/4 (c) 1/7 (d) 1/3
14. For KNN with L2 distance, training samples are sorted in: (a) descending order (b) ascending order (c) random order (d) class order

15. The Rocchio classifier assigns a new sample to the class whose: (a) nearest single sample is closest (b) centroid is nearest (c) prior is largest (d) variance is smallest

Answer key

Q	Answer	Q	Answer	Q	Answer
1	b	6	c	11	b
2	c	7	c	12	b
3	c	8	b	13	c
4	b	9	d	14	b
5	c	10	a	15	b